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The Editors-in-Chief
Brain Informatics
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Dear Editors,

We submit the research article **"What Does EEG-to-fMRI Synthesis Add to EEG-Based Diagnosis? A Controlled Benchmark in Psychosis, Schizophrenia and Parkinson's Disease"** for consideration in *Brain Informatics*.

EEG-to-fMRI synthesis, which predicts fMRI volumes from scalp EEG, is an active area of brain informatics, and several recent methods motivate it by its potential to support diagnosis where fMRI is unavailable. Whether synthesized fMRI adds diagnostic information beyond the EEG it is computed from has not, to our knowledge, been tested with the controls such a claim requires. We introduce a controlled evaluation protocol for this question and a benchmark that applies it. Four synthesizers (our earlier CNN-TAG model and the published E2fNet, Spec2VolCAMU-Net and NT-ViT) are trained on two public simultaneous EEG-fMRI datasets, with scanner-trigger alignment, and applied to resting EEG from five public clinical cohorts: two first-episode psychosis samples, chronic schizophrenia, Parkinson's disease and a schizophrenia-spectrum cohort recorded during a task. Synthesized fMRI is compared, alone and combined, with EEG band power and with an EEG view matched in information to the synthesizer input, under nested repeated cross-validation, permutation tests and corrected paired bootstrap; untrained (random-weight) synthesizers separate the learned mapping from the architecture, and artefact handling (ICA), covariates and recording conditions are analysed explicitly.

The main findings are as follows. No trained synthesizer added diagnostic information to the information-matched EEG view in any cohort, and the synthesized views that discriminated patients from controls, most clearly in Parkinson's disease, were matched by untrained networks. The one gain that survived correction came from an untrained network and disappeared after ocular artefacts were removed, while the Parkinson's disease discrimination was robust to artefact removal. The synthesizers did not outperform an EEG-free template image, and EEG explained a negligible part of the BOLD signal. Along the way, the protocol exposed recording-condition differences between groups in public clinical EEG releases (ocular activity, time of day, recording year, auxiliary channel layout) and alignment pitfalls in the most used simultaneous EEG-fMRI datasets, which matter for anyone using these data. We believe the protocol, the checklist of controls it yields and the released code will be useful to researchers developing cross-modal neuroimaging methods and to those considering them for clinical use. This fits the journal's scope at the intersection of brain science, informatics and medical decision support.

The manuscript is original and has not been published elsewhere, and it is not under consideration by any other journal. An earlier version of the synthesis-to-classification approach was posted as a preprint on medRxiv (doi:10.1101/2023.08.07.23293748 [VERIFY: DOI]); the present manuscript reports new experiments, cohorts and analyses. All analysed datasets are public, and all code is available at `https://github.com/DCalhas/eeg_to_fmri`. All authors have read and approved the manuscript and agree with its submission. The authors declare no

competing interests.

Yours sincerely,

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